Transcript: Cities & Climate Change

Cities: The Urban Environment

Slide 2: Video Topics

- ➢ Energy
- ➢ Emissions
- ➢ Land Use
- ➢ Vulnerability

In a statement on climate action, the United Nations wrote "Cities are major contributors to climate change. According to UN Habitat, cities consume 78 percent of the world's energy and produce more than 60 percent of greenhouse gas emissions. Yet, they account for less than 2 percent of the Earth's surface."

In this lecture, I am going to give consideration to that statement and look at cities and energy use, cities and greenhouse gas emissions, and cities, as a land use and also a factor in land use change.

I am also going to talk briefly about the impacts of climate change on cities and why they are some of the most vulnerable landscapes when it comes to a changing climate.

Slide 3: Cities & Energy

- ➢ Cities account for roughly three-quarters of the world's energy use
    - ○ also 80% of the global GDP and a majority of the population
- ➢ Buildings are major consumers of energy in cities (~ 40% to 80%)
    - ○ transportation and industry are the two other big consumers
- ➢ Electricity matters – fossil fuels (natural gas) are still the top source of electric power generation
    - ○ in the United States, natural gas is the top source used for electric power generation
    - ○ globally, the top source of electric power is coal

Cities occupy less than 2% of total land on Earth but consume far more energy than their rural counterparts. That said, they are home to a sizable and growing population, and more than three-quarters of economic activity and productivity on Earth is concentrated in cities.

Since energy is both the cause of and solution to climate change, cities present a great opportunity to reduce dependence on fossil fuels.

Apart from being centers of innovation, much of the energy used in cities is used in buildings for heating, cooling, and lighting, and powering appliances and electronics.

Transportation, a sector that relies almost wholly on petroleum is a big user, but not the biggest, user. And industry, a sector that relies largely on natural gas but also petroleum, is a user as well.

What this means is that electricity, or rather, the sources of energy used to generate electricity, matter. Globally, fossil fuels remain the top source, but in regions where fossil fuels, typically natural gas and coal, still command a significant share of the energy profile for electric power generation, there exists an opportunity to increase the use of renewable sources of energy like wind and solar, and, today, both are contributing more and more to electric power profiles.

In places where renewables are already contributing a large share, countries like Brazil, Canada, Sweden, Costa Rica, and Germany, this is positive, and hopefully, their share in the power generation mix will only increase. In most of these countries, hydroelectric power is an important contributor but the use of other renewable sources is increasing.

To summarize, most of the world’s energy consumption happens in urban areas. Buildings are the number one consumers of energy in cities, and their consumption varies pretty widely, between 40 and 80% of all consumption. Since buildings rely on electricity as their main source of energy, the potential is there to increase the use of renewable energy. That said, fossil fuels are still the top source of energy for electric power generation, and there are other users, like transportation and industry, using their fair share of fossil fuels as well.

Slide 4: Graph “Global energy consumption driven by more electricity in residential, commercial buildings”

Buildings, both residential and commercial structures, used about 20% of the world's total energy in 2018, the year this graph represents.

The age of the graph is not so much important as what it is illustrating. Natural gas, petroleum, and coal are still important sources of energy for buildings, but it is demand for electricity that is expected to increase.

The world’s urban population is growing and countries across Africa and Asia in particular are urbanizing rapidly creating a greater demand for energy to power buildings and electricity seems to be the preferred source of energy in non-OECD countries.

Non-OECD is basically, anywhere outside of North America, Western Europe, Japan, Australia, and New Zealand. There are a few other OECD countries, but that is most of them.

Data shows that there is a positive correlation between urbanization rate and electricity consumption. Electricity is the main energy source for things like lighting, cooling, appliances, and other electronic equipment, and it is the fastest-growing energy source in residential and commercial buildings.

In places where the urban population is growing rapidly, low-energy strategies for building design are critical.

Slide 5: Graphs “Global energy consumption driven by more electricity in residential, commercial buildings”

The U.S. Energy Information Administration predicts that global energy consumption in buildings will increase by an average of 1.3% per year by 2050, and non-OECD countries are expected to have a higher rate of energy consumption growth in buildings at more than 2% per year, or about five times the rate of OECD countries.

Historically, buildings in OECD countries have consumed more energy, but that is changing and consumption in non-OECD countries is on the rise. As is typically the case, non-OECD countries consume more electricity in total, but per capita consumption remains highest in OECD countries.

If we were to look at individual countries, China will have the largest increase in building electricity use in absolute terms, but India will experience the fastest growth rate in the decades to come. It is expected that China's buildings are expected to account for more than 20% of global building sector electricity consumption by 2050.

Between 2020 and 2050 it is also projected that electricity's share of total energy use in buildings will nearly double in non-OECD countries.

Slide 6: Image of the Chemistry Building at Michigan State.

If you check out MSU's sustainable facilities page you will learn that in 2009, MSU received its first Leadership in Energy and Environmental Design or LEED certification for renovations on the Chemistry Building. Since that time, several other buildings have also earned LEED certification and MSU ensures that all new construction and major renovation projects follow LEED standards. LEED standards are not just for energy efficiency and water conservation, they also concern how materials are used, what materials are used, and what happens to the waste, from start to finish.

The technology to construct sustainable buildings exists, but at least upfront, it can be costly and take more time. Some strategies, like passive solar design, low flow plumbing fixtures, smart thermostats, occupancy sensors for lights, insulation, and seals, all can be fairly inexpensive to install and save money. Some of these things are the norm for new construction.

Green roofs, or vegetated roofs, are also a fairly simple strategy for making buildings more energy efficient. They have several benefits, but what is probably their greatest benefit is that they can help reduce energy use in buildings. They provide shade and act as a thermal mass buffer or insulator of sorts, which can reduce the demand for heating in cool seasons or in the evenings and reduce the demand for cooling in warm seasons or at warm times of the day.

Slide 7: Cities & Greenhouse Gas Emissions

- ➢ Cities account for 60 to 70% of global greenhouse gas emissions
  - there is a positive correlation between urbanization rate and electricity consumption
    - electric power generation, transportation, and industry are the three largest contributors to global emissions and climate change
    - buildings and transportation directly account for most of a city's emissions
    - waste generated in cities also has a greenhouse gas emissions component as do the building materials used in cities

Energy use is directly related to greenhouse gas emissions and so much of what is on this slide is a reflection of what we have already talked about.

I also want to note that it is immensely difficult to quantify greenhouse gas emissions from cities because of the lack of directly measured emissions data. Carbon dioxide emissions estimates are available for fewer than one hundred cities worldwide, about half of which are in low- and lower-middle-income countries. And most of these estimates are not based on ancillary data, not direct measurements. The use of satellite-based CO_2 measurements is up and coming and has the potential to really expand the scope of emissions-performance ratings.

With that said...

We've established that electricity, natural gas, and oil are the dominant energy sources used in cities, and all contribute to greenhouse gas emissions. There is enormous potential in decarbonizing electricity generation, but at this time, if you look at carbon dioxide emissions by end-use sector, electric power generation is a major source of emissions and a contributor to climate change.

This is particularly true in coal-dependent countries like China, Australia, India, and South Africa. And even though the United States is more reliant on natural gas, also a fossil fuel, a significant amount of emissions still comes from coal-fired electric power plants.

Coal combustion is more carbon-intensive than burning natural gas for electric power production and even in the United States, coal use accounts for more than half of the carbon dioxide emissions from the electric power sector.

We know that there is potential for a more carbon-neutral electric power sector and cities, in general, but there are many countries with large and growing urban populations that have planned coal-fired power plant projects in the works including Nigeria, Kenya, Tanzania, India, and China.

I've listed some other sources of carbon dioxide emissions in cities, including transportation and industry, with cars, trucks, and buildings being directly responsible for emissions, alongside industrial activities. Waste and the manufacture of construction materials like concrete and steel are also sources.

Slide 8: Graph "Greenhouse gas emissions by sector, World"

This chart shows the breakdown of greenhouse gas emissions measured in tonnes of carbon dioxide equivalents by sector. Globally, electricity and heat production are the largest contributors to greenhouse gas emissions, followed by transport, manufacturing and construction, and agriculture.

The contribution of these sectors does vary from country to country. In the United States, transport has a bigger impact on emissions than the global average, while in Brazil, a larger share of emissions come from agriculture and land use change.

Cities are concentrations of industry, buildings, transportation, manufacturing, construction, electricity, and heat, it only makes sense that they would be concentrations of greenhouse gas emissions.

Slide 9: Graph "Greenhouse gas emissions by sector"

Energy, which powers buildings, transport, and industry, along with some other miscellaneous things, accounts for a lion's share of greenhouse gas emissions.

The proportion of that coming from buildings is 17.5%, 10.9% from residential buildings and 6.6% from commercial buildings. Transport and industry are the two other contributing sectors that all combined make up that 73.2% you see assigned to energy.

I am going to talk a little more about the construction materials that build cities in a moment, but for now, understand that cities consume copious amounts of iron and steel, wood, and cement, all of which have associated greenhouse gas emissions.

Earlier I noted that cities are responsible for 60% of greenhouse gas emissions globally, which was the United Nations Habitat estimate. The World bank estimate puts that number above 70%, and you can see why. And when I give you these numbers, I'm not really interested in you knowing specific numbers, only understanding magnitude–and it's a lot.

Graphs can be somewhat confusing because electricity generation as a sector has associated emissions and buildings, which use a lot of electricity, also have associated emissions because they rely not only electricity but other fuels as well, mostly natural gas for heating.

Slide 10: Cities & Land Use Change

- Urban heat islands
  - man-made building materials conduct, absorb, and reflect heat differently than natural vegetation
  - waste heat is generated by machines (such as cars and HVAC systems)
  - evaporation from the surface of vegetation has a cooling effect
    - vegetation is also a source of shade

With energy and emissions going hand in hand, I feel like I am finally talking about something a little different and that is urban heat islands and how the change in land use from natural vegetation to man-made construction materials and the introduction of human activities contributes to a rise in temperatures.

Known as the urban heat island effect, urban areas experience higher temperatures than surrounding rural areas due to the absorption and retention of heat by buildings, pavement, and other infrastructure, as well as the lack of shading and cooling vegetation, and the addition of waste heat.

Cities consist of many man-made surfaces: concrete sidewalks, steel buildings, and asphalt streets all come to mind. In rural environments, like parks in a city, vegetation tends to dominate the landscape.

The differences in the way the materials, man-made and natural vegetation, conduct, absorb, and reflect heat energy is the primary reason for the temperature discrepancy. Evaporative cooling, which occurs when water evaporates from plant and soil surfaces, is another reason for the difference.

Lastly, waste heat generated by automobiles and other machines also contributes to the warming effect in an urban core.

Combined, these land cover and land use differences can translate to a mean annual temperature difference anywhere from two to ten degrees Fahrenheit during the day, and even more at night.

Among other things, this difference can increase cooling needs, degrade air and water quality, and impair annual photosynthetic productivity in nearby vegetation.

Slide 11: Images “The Making (and Breaking) of an Urban Heat Island”

These are satellite photos of New York, NY showing the difference vegetation can make on surface temperatures within a city.

It reads: “On a hot, sunny afternoon, walk outside and find a parking lot bordered by grass. Put one hand on the asphalt and the other on the grass. The same surfaces of asphalt, stone, brick, and cement that keep weeds and water out of our way can sizzle in the sunlight and raise local temperatures”. To make room for all these buildings and roads, cities squeeze out vegetation that would otherwise cool its surroundings by evaporating water. Added to the mix are car motors, hot air from air conditioners and clothes dryers, earth-moving machines, even smokestacks. All of these heat sources work together to raise temperatures.

The urban heat island effect is perhaps most dangerous during heat waves, when demand for air conditioning soars…

This Earth Observatory feature was reporting a study investigating the degree to which lighter surfaces, urban forestry, and green roofs could cool the city during the hottest times of the year.

Researchers found that reducing dark surfaces found covering more than 60% of the city and the addition of green roofs had the greatest impact and could cool the city by about two to three degrees Fahrenheit.

While that might not seem like a lot, it would go a long way toward reducing the demand for electricity during the hottest times of the year.

Slide 12: Materials Consumption and the Projected Change in Amount of Domestic Materials Consumption

Man-made surfaces absorb more heat and heat the surrounding air, but there is also a significant energy consumption and greenhouse gas emissions component associated with the demand for manufactured construction materials like steel and cement and mined materials like sand.

Earlier I put up a pie chart that included greenhouse gas emissions associated with energy consumption, and part of that was the manufacture of iron and steel and cement and energy-related emissions associated with other industries, including mining and quarrying. More than 10% of global emissions come from activities directly related to the manufacture of construction materials.

Visual Capitalist did a feature on materials consumption and the change in the amount of domestic materials consumption (DMC) by world region that is expected to occur in the 40 year period between 2010 and 2050. I will spare you the details of how this is measured but there were a few basic conclusions.

First, material consumption is highly uneven across the different world regions with the world's wealthiest countries consuming 10 times as much as the poorest, and twice the global average.

Based on the total urban DMC, East Asia leads the world in material consumption, with China consuming more than half of the world's aluminum and concrete.

The biggest jump in the next decades will occur in Africa, where the population is expected to almost double, and material consumption will jump from just 2 billion tonnes in 2010 to 17.7 billion tonnes per year.

Major Global Regions	**2010 Material Consumption** (billion tonnes)	**2050P Material Consumption** (billion tonnes)	**% total change** (2010-2050P)
Africa	2.0	17.7	792%
Southern Asia	2.7	8.6	223%
South-Eastern Asia	2.0	5.6	180%
Central and Western Asia	1.9	4.7	151%
Oceania	1.1	2.6	136%
Eastern Asia	9.0	19.2	113%
South and Central America	6.5	11.1	71%
Europe	8.3	10.4	25%
North America	7.7	9.0	17%
World	**41.1**	**88.8**	**116%**

Slide 13: Cities & Vulnerability

- ➢ Climate change increases the frequency and severity of weather-related hazard events
  - the presence of people and their human landscapes (cities) increase the likelihood that a hazard is a disaster
  - many cities have heightened physical and economic vulnerability
  - weather-related hazards (floods, cyclones, heatwaves) pose a threat to public health, water resources, built landscape, energy security, and more
- ➢ Sea level rise threatens coastal cities

Natural hazards are defined as recurring natural phenomena that pose a threat or have the potential to cause harm. These events can and do occur at every location on Earth and they may be atmospheric, hydrological, geologic, or biological in nature.

Hazards are naturally occurring events, disasters are not. The word disaster introduces the human element of hazards. Disaster means that human lives and property are being threatened.

Climate change increases the likelihood that weather-related hazards will happen and that they will be severe. Cities increase the likelihood that a disaster will occur due to their relatively high population density and concentration of buildings and infrastructure.

Vulnerability is the predisposition to be hurt by an event beyond a certain magnitude and there are four types of vulnerability, one of them being physical. Physical vulnerability means that certain characteristics of the physical location make an area more vulnerable when a cyclone or hurricane makes landfall, or a heatwave sets in.

Three factors make a location more susceptible to harm: 1) the geography of the area, so its topography, setting, and accessibility, 2) the number of people living there; and 3) the population density.

The accumulation of wealth in cities, in terms of land, labor, and capital, and the great degree of socio-economic inequality also means that they are economically vulnerable, and likely to suffer greater economic losses. Further, lower-income groups in cities are less able to plan for and absorb the financial loss associated with a disaster

Weather-related hazards associated with global climate change pose a severe threat to human lives and their overall well-being and cause damage to the built landscape, including infrastructure and buildings. And this threat is greatest in cities due to their physical and economic vulnerability.

Climate change and weather-related events can impact the economy of cities, disrupting tourism, transportation, and other industries.

And an increased prevalence of droughts and heatwaves will impact the availability of water for cities and can also exacerbate health issues such as respiratory illnesses, allergies, and heat-related illnesses in urban areas. More frequent and severe heat waves will also lead to more blackouts and a less stable energy supply when temperatures rise and the demand for electricity rises.

Cities are also particularly vulnerable to the effects of sea-level rise, which can cause flooding and damage to buildings and infrastructure.

Slide 14: Image of the Sea Level Rise Viewer

In the United States and worldwide, coastlines are home to 40% of the population and that number is growing. 14 of the world's 17 largest cities are located on a coastline and on coastlines, the population density is double the global average.

This means that we have a ton of people living in coastal zones, concentrated on a relatively small amount of land, and physically vulnerable to rising sea levels, tides, storm surges, and other coastal hazards.

The coastal environment is increasingly urban and the population living there is growing. A result of climate change, melting glaciers and ice sheets, and warming ocean waters are causing sea levels to rise, which is intensifying the risk of flooding, erosion, and saltwater intrusion to coastal cities.

Scientists can't say with certainty just how much higher they will be in the future, but they are rising and this rise is expected to accelerate in the coming decades.

This is a map of the United States showing vulnerable coastal areas. The darkest red areas are the most vulnerable and the blue represents projected water depth. With NOAA's interactive sea level rise viewer, you can adjust the “mean higher high water” level to see how vulnerability changes. This is at current “mean higher high water”.

São Paulo, Brazil, Bangkok, Thailand, Jakarta, Indonesia, Lagos Nigeria, Shanghai, Tokyo, New York ...all of these cities and more are on coastlines.

Slide 15: Image of Mangroves

Mangroves are coastal forests of salt-tolerant trees and shrubs that grow in shallow tidal waters of the tropics. Mangroves are found along 25% of the world’s tropical coastlines and they play a critical role in protecting coastal areas, including cities, from flooding and storms. Mangroves also store a lot of carbon in their soils and biomass.

When Mangroves are destroyed, that carbon enters the atmosphere, contributing to greenhouse gas emissions and climate change. The coast also loses its natural barrier against storms, sea level rise, and flooding. Their role in marine ecosystems is invaluable as well, but for the purpose of this lecture, I will leave it at that.

In some countries, the loss of mangroves has been significant. Kenya, Liberia, the Philippines, and Puerto Rico have all lost over 70% of their mangrove forests. A study by the World Resources Institute concluded that worldwide, mangrove loss was strongly correlated with the growth of cities and ports. In turn, the loss of mangroves, and environmental degradation in general, increases the vulnerability of cities. Environmental degradation being an aspect of environmental vulnerability.

Slide 16: Concluding Thoughts

- ➢ Urbanization and climate change go hand in hand, with cities contributing to climate change, while also being some of the most vulnerable locations to the impacts of climate change.
- ➢ As urban populations continue to grow, cities will play an increasingly important role in global efforts to reduce greenhouse gas emissions and adapt to the impacts of climate change.

Urbanization and climate change go hand in hand, with cities contributing to climate change, while also being some of the most vulnerable locations to the impacts of climate change.

As urban populations continue to grow, cities will play an increasingly important role in global efforts to reduce greenhouse gas emissions and adapt to the impacts of climate change.

Addressing the impacts of urbanization on climate change will require a combination of policy, technology, and behavioral changes. Governments, businesses, and individuals can all take steps to increase energy efficiency, decrease their reliance on fossil fuels, and ultimately reduce emissions from transportation, buildings, and industry.

Investing in more sustainable infrastructure, energy-efficient buildings, and green spaces that can help mitigate the impacts of climate change, could go a long way.

On the flip side, cities will need to have a plan and place to decrease their vulnerability to a changing climate and related impacts like severe weather events and rising sea levels. They will also need to increase their resilience and ability to recover, as these things are already happening and will continue to happen with greater frequency and magnitude.

One thing that I did not have time to get into is the role that cities, and their mayors, are playing in addressing climate change. Mayors and city leaders around the world are setting emissions reduction targets, investing in sustainable infrastructure and public transit, incentivizing renewable energy and energy efficiency, and leading initiatives to help their cities adapt to climate change.

And with that, I will conclude this lecture on cities and climate change.